(502) 472-5677 **Karthik Nayak** **nayakkarthik02@gmail.com**

Education

Bachelor of Science in Computer Science — University of Kentucky
GPA: 3.6 / 4.0, Junior — Projected graduation: May 19

Skills

Programming	Technologies	Libraries	Cloud Services	Front-End	Concepts
Python	Docker	OpenCV	AWS	JavaScript	Machine Learning
Java	Build Automation	TensorFlow	Azure	Angular	Image Processing
C++	Continuous Integration	NumPy	Google Cloud Platform	HTML/CSS	API Design
C	Virtualenv	Flask		Graphic Design	OOP

GitHub: /karthiknayak02 **Website**: karthiknayak02.github.io

Work Experience

IBM *June. 2018 – Aug. 2018*
Software Developer Intern *Dallas, TX*

- Worked on IBM's BTI supply chain SaaS product to convert existing monolithic code to microservice architecture
- Designed and developed several REST API's in JAVA. Integrated implementation using cloud docker containers
- Used technologies like Maven for build automation, Jenkins for continuous integration and Git for version control
- Built dynamic and responsive UI using Angular to consume and test our microservice data

Tesla *May. 2017 – Aug. 2017*
Automation Controls Engineering Intern *Gigafactory 1, Reno, NV*

- Worked in the Model 3 Controls Team to support software design and programming of several production machines
- Developed PLC Add-on Instruction (AOI) packages; for interactions between (AGV) Automated Guided Vehicles - workstation conveyers. and other machines on M3 drive unit line
- Built a simulator to identify and correct programming flaws in AGV equipment throughout Gigafactory1
- Designed various user interface (UI) models and integrated PLC programs on the drive unit powertrain

Visumatic Industrial Product *May. 2016 – Aug. 2016*
Computer & Electrical Engineering Intern *Lexington, KY*

Visumatic is an OEM of fastener "feed and drive" equipment and automated systems.

- Worked in the development and implementation of automated systems (robots) for manufacturing
- Supported the calibration and build of 6-Axis robot systems; helped configure vision systems, OS, troubleshoot electro-pneumatically controlled mechanical feeders and drivers and verify electrical schematics
- Designed and developed program logic and UI for various machines with pneumatic/DC drivers, touch HMI displays, rotatory indexing systems from the ground up and also authored the logic manual for the machine

Part-Time

University of Kentucky, Tau Beta Pi *Aug. 2018 – Current*
Computer Science Tutor *Lexington, KY*

- Assisting fellow undergraduates in computer science courses through Tau Beta Pi's Peer Tutoring Program

University of Kentucky, Computer Science dept. *Oct. 2017 – May. 2018*
Undergraduate Research Assistant *Lexington, KY*

- NSF-funded research project to study Software-Defined Networks (SDN) on campus network architectures
- Developed a client/server storage application using Python along with Flask web framework
- Assessed speed tests of VIP internet lanes using rclone - automated using shell scripts

Achievements

North America IBM Intern Hackathon - Team Lead *2018*
- Developed a analytics solution with Watson Vision to perform video analysis on gym equipment use (Team of 3)

2nd Place, Cathacks IV national college hackathon - Team Lead (Major League Hacking) *2018*
- Developed a Chrome Extension which applies Natural Language Processing (NLP) on video captions to extract topics in the video. When clicked uses code injection to automate video scrubbing to the corresponding section of the video (Team of 4)

2nd Place, Cathacks III & winner of AWS challenge (Major League Hacking) *2017*
- Developed an Amazon Alexa application with AWS to get user speech input and generate 3D Braille STL files (Team of 2)

Eta Kappa Nu – ECE Honor Society for students in upper 25% of the junior class 2017